

Mitchell Piehl

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Summary

AI/ML researcher and developer specializing in LLM/MLLM memory, agentic systems, RAG, and evaluation with published, state-of-the-art research and production AI deployed across an enterprise.

Education

UNIVERSITY of Iowa

Iowa City, IA

Ph.D. Computer Science | Sheets Fellowship Recipient | Advisor: Dr. Muchao Ye

August 2025 - Present

Focus: Memory mechanisms in LLMs/MLLMs, agentic systems, AI reasoning, Computer Vision, and NLP

UNIVERSITY of St. Thomas

St. Paul, MN

B.S. Computer Science | 3.98 GPA

May 2025

Minors: Data Science, Applied Statistics, Philosophy and Science, Logic & Analytical Reasoning

- D1 Track and Field Captain | All-Conference Athlete | 7x School Record Holder | SAAC Representative

Technical Skills

Languages/Tools: Python, PyTorch, TensorFlow, HuggingFace, scikit-learn, Pandas, NumPy, Pydantic

LLMs: Open- and closed-source APIs, RAG, agent design, multimodal modeling, test-time learning

Expertise: LLM Memory, LLM Evaluation & Benchmarking, Agentic AI, GenAI, NLP, Reasoning in AI, CV

Research & Professional Experience

Inspire Medical Systems, Inc.

Minneapolis, Minnesota

AI Development Intern

May 2026 – Present

- Built and deployed an internal web application for an LLM-powered document processing tool, reducing processing time by ~80% across 12+ internal teams.
- Developed RAG AI agents to automate tasks across the organization, with rigorous evaluation
- Bridged research concepts into production-ready systems and presented findings to AI stakeholders.

University of Iowa

Iowa City, Iowa

Ph.D. Fellow and Graduate Researcher

August 2025 - Present

- Built MemFit (submitted ICLR 2027), a tree-based agentic memory system reaching state-of-the-art accuracy across many benchmarks while using 50% fewer LLM calls, 61% fewer tokens, and 4x-50x faster memory construction than prior systems.
- Designed ER-MIA (submitted EACL 2026), a black-box adversarial memory-injection framework using SVD embedding optimization that degraded LLM reasoning F1 by up to 65%, and proposed defenses against embedding-space attacks.
- Developed LATERN (submitted WACV 2026), a video anomaly detection pipeline that introduces an image-grounded memory mechanism to reformulate VAD as a temporal evidence aggregation process.
- Engineered end-to-end pipelines and multi-metric evaluation harnesses (F1, BLEU, ROUGE, Exact Match, LLM-as-judge) to benchmark memory and RAG systems.

University of Colorado Colorado Springs

Colorado Springs, CO

Natural Language Processing Researcher

May 2024 – August 2024

- First-author paper accepted to IEEE ICMLA 2025; designed EVoSS, an LLM math-reasoning method combining equation generation with estimation-based verification for state-of-the-art results (~2% average benchmark gain).
- Built new and improved existing benchmark datasets to evaluate LLM mathematical reasoning.

Publications

Solving Math Word Problems Using Estimation Verification and Equation Generation (EVoSS)

- Published IEEE ICMLA 2025 (first author). <https://arxiv.org/abs/2509.18565>

ER-MIA: Black-Box AMIAS on Long-Term Memory-Augmented LLMs

- Submitted EACL 2026 (first author). <https://arxiv.org/pdf/2602.15344>

MemFit: Efficient Long-Term Agentic Memory

- Submitted NeurIPS 2026 (first author).

LATERN: Test-Time Context-Aware Explainable Video Anomaly Detection

- Submitted WACV 2026 (first author). <https://arxiv.org/pdf/2605.15054>